

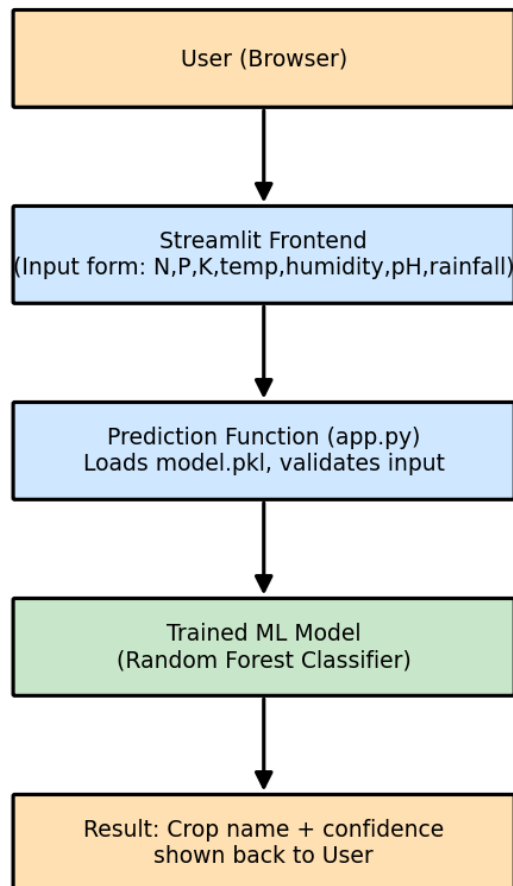
Project Design Phase

Solution Architecture

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Architecture Diagram:

OptiCrop - Solution Architecture



Components Description:

S.No.	Component	Role
1.	Streamlit Frontend	Displays the input form (7 fields: N, P, K, temperature, humidity, pH, rainfall) and shows the prediction result to the user
2.	app.py (Backend Logic)	Validates user input, loads the trained model, calls model.predict(), handles errors gracefully
3.	model.pkl (Trained ML Model)	Pre-trained Random Forest Classifier, loaded at runtime, returns predicted crop and confidence score
4.	Crop Recommendation Dataset	Used only during model training (offline), contains 2200 records with 7 features and 22 crop labels
5.	Streamlit Community Cloud	Hosts the deployed web application publicly, connected directly to the GitHub repository

Deployment Architecture:

The app.py file and model.pkl are pushed to a public GitHub repository. Streamlit Community Cloud connects to this repository and automatically deploys and serves the web application at a public URL. Any update pushed to GitHub triggers an automatic redeploy.